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Importance of C

- It is a robust language whose rich set of built-in functions & operators can be used to write any complex program.
- Programs written in C are fast & efficient. This is due to its variety of data types and powerful operators.
- C is highly portable and machine independent language. C programs written for one computer can be run on another with little or no modification.
- In C, several standard built-in functions are available which can be used for developing programs.
- The C compiler combines the capabilities of an assembly language with the features of a high-level language and therefore it is well suited for writing both system software & business packages.

C is a programming language developed at AT & T's Bell Laboratories of USA in 1972. It was designed and written by Dennis Ritchie.

Keywords, Constants & Variables in C

- **Keywords** :- Keywords are the words which have special meaning and whose meaning has already been explained to the C compiler. The keywords cannot be used ~~to~~ as variable names because if we do so, we are trying to assign a new meaning to the keyword, which is not allowed. This is why, keywords are also called reserved words.

Examples are auto, int, char, float, case, break, etc.

Constants are values that do not change during the entire program execution.

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- **Variables** :- A variable is an entity that may change. In a C program, we do many calculations. The results of these calculations are stored in computer's memory cells which are also called memory locations. To make the retrieval and usage of these values easy, ~~these~~ names are given to these memory locations. Since the value stored in each location may change, the names given to these locations are called variables.

- **Rules for constructing variable names**

1) A variable name is any combination of alphabets, digits or underscores.
2) The first character in a variable name must be an alphabet or underscore.
3) No commas or blanks are allowed within a variable name.
4) No special symbol other than an underscore can be used in a variable name.

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C Instructions:

There are 3 types of instructions in C :-

1) Type Declaration Instruction – This instruction is used to declare the type of variables used in a C program.

2) Arithmetic Instruction – This instruction is used to perform arithmetic operations on constants and variables.

3) Control Instruction – This instruction is used to control the sequence of execution of various statements in a C program.

- Hierarchy of Operations

% → Modular division

Priority	Operators
1st	*, /, %
2nd	+, –
3rd	= (Assignment)

- Control Instructions in C

There are 4 types of control instructions in C:-

1) Sequence Control Instruction:- It ensures that the instructions are executed in the same order in which they appear in the program.

2) Selection or Decision Control Instruction
3) Case Control Instruction

These instructions allow the computer to take a decision as to which instruction is to be executed next.

4) Repetition or Loop Control Instruction :-

These instructions are used to execute a group of statements repeatedly.

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Operators in C

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An operator is a symbol that tells the computer to perform certain mathematical or logical manipulations. Operators are used in programs to manipulate data and variables.

Operators in C can be classified as follows:-

1) Arithmetic Operators
2) Relational Operators
3) Logical Operators
4) Conditional Operators / Ternary Operators
5) Increment & Decrement Operators
6) Assignment Operators
7) Bitwise Operators
8) Special Operators

1) Arithmetic Operators

These are as follows:-

Operator	Meaning	Example
+	Addition	10 + 5 = 15
-	Subtraction	10 - 5 = 5
*	Multiplication	10 * 5 = 50
/	Division	10 / 5 = 2
%	Modulo division	14 % 5 = 4

2) Relational Operators

We often compare two quantities and depending on their relation, take certain decisions. These comparisons can be done with the help of relational operators. An expression containing a relational operator is termed as a relational

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expression. The value of a relational expression is one if the specified relation is true and zero if the relation is false.

Relational Operators	Meaning
<	is less than
<=	less than or equal to
>	greater than
>=	greater than or equal to
==	is equal to
!=	not equal to

Logical Operators

In C, there are 3 logical operators:-

1) AND (&&)
2) OR (||)
3) NOT (!)

The first 2 operators (&& and ||) allow two or more conditions to be combined in an if statement. These operators are used when we want to test more than one condition and make decisions.

For example, the logical expression (a>b && x==5) is true only if both a>b is true and x==5 is true. If either of them is false, the expression is false. On the other hand, the logical expression (a>b || x==5) is true if either of a>b is true or x==5 is true. If both of them are false, only then the expression is false.

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Conditional Operators / Ternary Operators

The operators ? and : are called conditional operators. These are also called "ternary operators" since they take three arguments. They form a kind of foreshortened if-then-else. Their general form is :-

expression 1 ? expression 2 : expression 3

This expression means that "If expression 1 is true, then the value returned will be expression 2, otherwise the value returned will be expression 3".

Example:-

```
a = 20
b = 30
x = (a>b) ? a : b;
```

Here, x will be assigned the value of b. i.e. x = 30.

Increment and Decrement Operator

The increment operator ++ adds 1 to the operand and the decrement operator -- subtracts 1 from the operand.

Both are unary operators and take the following form:-

```
m++; or ++m;
m--; or --m;
```

m++ or ++m is equivalent to m=m+1; or m+=1;
& m-- or --m is equivalent to m=m-1; or m-=1;

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Assignment Operator (=) :-

It is used to assign the result of an expression to a variable.

Eg: $x = a + b$ means that the values of variables a and b are added and their sum is assigned to the variable x. So, if a and b initially had values 10 and 20 respectively, then their sum ie. 30 is assigned to x. So, now $x = 30$.

Bitwise Operators :-

To be able to access or manipulate individual bits, C language provides a set of bit manipulation operators. These are as follows:-

Operator	Meaning
~	One's complement
>>	Right shift
<<	Left shift
&	Bitwise AND
|	Bitwise OR
^	Bitwise XOR

These operators can operate on integers and characters but not on floats and doubles.

Special Operators :-

C supports some special operators such as:-

i) Comma Operator (,) :- It can be used to link

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related expressions together. A comma-linked list of ~~operators~~ expressions are evaluated left to right and the value of right-most expression is the value of the combined expression.

Eg:- the statement ⇒ value = (x=10, y=5, x+y);
at first assigns the value 10 to x, then assigns 5 to y and finally assigns 10+5 ie 15 to value.

ii) Size of Operator

~~It~~ This operator when used with an operand, returns the number of bytes the operand occupies. The operand may be a variable, a constant or a data type qualifier.

Eg:- Suppose sum is declared as an integer variable.
Then, m=size of (sum) will ~~return~~ assign the value 2 to m because an integer variable occupies 2 bytes in memory.

Similarly, n = size of (float) will assign the value 4 to n.

iii) Pointer operators (& and *)

'&' is C's 'address of' operator. The expression p=&a means the address of variable a is collected in a variable p.

The operator '*' stands for 'value at address'. So, int *n means the value at address contained in n is an integer.

iv) Member Selection operators (• and →)

These operators are used to select members of a structure. • operator is

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used to access a structure ~~member~~ element, ~~using~~ through a structure variable. Structure elements can be accessed through a pointer to a structure using the arrow (→) operator.

Precedence of Arithmetic Operators

An arithmetic expression without parentheses will be evaluated from left to right using the rules of precedence of operators.

There are 2 distinct priority levels of arithmetic operators in C :-

High priority → * / %

Low priority → + −

Let us consider the following statement:-

$$y = a + b/5 + c*9 - 1$$

Suppose, a = 10, b = 15, c = 3, then the statement y = 10 + 15/5 + 3*9 −1 is evaluated as follows:-

$$y = 10 + \underline{15/5} + 3*9 - 1$$

ie. $y = 10 + 3 + \underline{3*9} - 1$

ie. $y = \underline{10 + 3 + 27} - 1$

ie $y = 40 - 1$

ie. $y = 39$

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Size & Range of Data Typer on a 16-bit machine

Data Type	Size (byte)	Range
char	1 byte	-128 to 127
int	2 "	-32,768 to 32,767
float	4 "	3.4E-38 to 3.4E+38
long int	4 "	-2,147,483,648 to 2,147,483,647
double	8 "	1.7E-308 to 1.7E+308
long double	10 "	3.4E-4932 to 1.1E+4932

Note : On a **32-bit compiler**, size of int is 4 bytes, size of long double is 16 bytes.

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